Verification report

neuro-san-esp at 6c84a9b — produced by running every command below, 2026-08-23 00:59 UTC

Each check in this document was executed to produce the row that describes it. The exit code is the verdict. Nothing here is a summary of a previous run, and nothing is asserted that was not observed — a report that cannot record a failure is not evidence of a success.

Summary

Check	Exit	Result
Test suite	0	PASS
Lint	0	PASS
Offline search (no key, no network)	0	PASS
Preflight, with provider key	0	PASS
Champion network generation	0	PASS

Every executed check passed

That covers what can be run without a provider key. It does not cover the four claims on the last page, which this environment cannot test at all.

Test suite

Every property this project claims about itself, asserted.

$ python -m pytest tests -q — exit 0

```
........................................................................ [ 30%]
........................................................................ [ 60%]
........................................................................ [ 90%]
......................                                                   [100%]
238 passed in 13.42s
```

Lint

The same command CI runs, on the same paths.

$ python -m ruff check esp tests scripts apps — exit 0

```
All checks passed!
```

Offline search (no key, no network)

Phase B and C: the half of ESP that is meant to cost nothing.

$ python offline_search.py --pool 2000 — exit 0

```
No local measurements in .esp-cache -- falling back to the committed ones in tests/fixtures/cache.
Run `make baseline` with a key to train on your own instead.

Phase B -- training the Predictor on 3 real evaluations from tests/fixtures/cache
  surrogate on 3 samples: rank quality NOT MEASURED -- cross-validation needs 8. The predictor is untrained and retur

Phase C -- evolving 2000 candidates against it
  2000 viable, 822 rejected as invalid
  scored in 0.11s with zero provider calls
  ~0.057 ms per candidate

  !! The surrogate is UNTRAINED on 3 samples (needs 8). Every prediction below is the same constant, so this is NOT a

First 5 in generation order -- all tied, see above:
  +0.7842  915e149aed588d57  agents=5 depth=2  via split_agent
  +0.7842  0334802d6e9b1376  agents=2 depth=2  via merge_agents
  +0.7842  7edd0c338d311ac0  agents=5 depth=3  via add_agent
  +0.7842  9f1d2fa76d71da2e  agents=4 depth=2  via split_agent
  +0.7842  0334802d6e9b1376  agents=2 depth=2  via merge_agents

Phase C scored 2000 candidates in 0.1s. The same number evaluated for real, at roughly 8 minutes each, would take abou
Phase C generated and scored them for nothing, which is the cost claim. It did not rank them, and it never measures --
```

Preflight, with provider key

passes when a provider key is configured, refuses when one is not. Which of those is the pass condition depends on the environment this report was produced in, so it is decided by looking rather than assumed -- an earlier version of this file hardcoded 'refusal is the pass' and then reported a healthy preflight as a failure the moment a key was added. Here: a key was present, so exit 0 is the pass.

$ python run_optimizer.py --check — exit 0

```
[ok  ] provider key: set, GOOGLE_API_KEY, from .env
  [ok  ] AGENT_TOOL_PATH: /home/user/neuro-san-esp
  [ok  ] PYTHONPATH: /home/user/neuro-san-esp
  [ok  ] state directory writable: /home/user/neuro-san-esp/state
  [ok  ] designer demo mode: off
  [ok  ] model ladder: gemini-3.5-flash-lite, gemini-3.1-flash-lite -- 1000 requests/day = about 6 candidate(s)
```

Champion network generation

Writes the best measured topology into the registry as a servable agent.

$ python serve_champion.py --state state — exit 0

```
No service population yet -- serving the best topology from the committed measurements in results/history.json.
Run a wake with a key to serve your own.

champion: designer_shaped (459ac1a66d925b0c)
  accuracy 0.82   tokens 278,532   agents 4   fitness +0.7868
  wrote registries/champion.hocon
  wrote registries/champion_manifest.hocon

Serve it:
  AGENT_MANIFEST_FILE=/home/user/neuro-san-esp/registries/champion_manifest.hocon \
  AGENT_TOOL_PATH=/home/user/neuro-san-esp PYTHONPATH=/home/user/neuro-san-esp \
  python -m neuro_san.service.main_loop.server_main_loop
```

Not verified here

Four claims this environment cannot test. Named, not omitted.

These are the rows that matter most in a document like this. A check that could not run is not a check that passed, and the distance between those two is where a project starts claiming more than it measured.

An agent answering a question

Needs a provider key, and this session has none. The agent networks render, the server accepts them and the corpus tool returns documents, but nothing has called a language model, so no answer has been produced or scored here.

OpenRouter as the provider

The switch is implemented and its configuration is tested, but openrouter.ai is refused by this sandbox's egress policy (HTTP 403 on CONNECT), so no call has been made to it. Google's endpoint is reachable from here; OpenRouter's is not.

The container image

No Docker daemon in this environment. tests/test_container.py parses the Dockerfile and compose.yaml instead and checks that everything the documented commands need is copied. That is file inspection, not a build.

A full evolutionary run

One candidate costs about 165 provider requests and the free tier caps requests per day per model. No search has completed, and no evolved candidate has beaten the seed baseline.

What would close the gap

A free Google AI Studio key, pasted into `.env`. Google's endpoint is reachable from the environment this report was produced in, so the preflight, a real wake, and an agent answering a question would all become testable. One candidate costs about 165 requests against a free tier that allows 500 per day per model, so the first three measurements are affordable on day one and the surrogate's eight-sample floor is not.